



Ansible Cheatsheet

Complete Quick Reference Guide for Ansible Automation



350+ Commands



500+ Examples



Production Ready

EngiD Dock

Basics & Installation

| Installation & Setup

Install Ansible:

```
pip install ansible
pip install ansible==2.10.7      # Specific version
apt-get install ansible          # Ubuntu/Debian
yum install ansible              # RHEL/CentOS
brew install ansible             # macOS
```

Check Ansible version:

```
ansible --version
ansible --list-hosts inventory.ini
ansible -m setup localhost | head -20
```

| Inventory File

Basic inventory format:

```
[webservers]
web1.example.com
web2.example.com ansible_user=ubuntu

[databases]
db1.example.com ansible_port=2222
db2.example.com

[all:vars]
ansible_python_interpreter=/usr/bin/python3
ansible_ssh_private_key_file=~/.ssh/id_rsa
```

Group variables:

```
ansible_host=192.168.1.1
ansible_port=22
ansible_user=root
ansible_password=secret
ansible_ssh_private_key_file=/path/to/key
ansible_connection=ssh
```



Ad-Hoc Commands

Running Commands

Execute module:

```
ansible all -i inventory -m ping
ansible webserver -m shell -a "uptime"
ansible all -m command -a "ls -la /tmp"
ansible databases -m shell -a "mysql -v"
```

Common modules:

```
ansible all -m debug -a "msg='Hello'"
ansible all -m copy -a "src=/tmp/file dest=/tmp"
ansible all -m file -a "path=/tmp state=directory"
ansible all -m service -a "name=nginx state=started"
ansible all -m user -a "name=john state=present"
```



Playbooks

Playbook Structure

Basic playbook:

```
---
- name: Deploy application
  hosts: webserver
  vars:
    app_port: 8080

  tasks:
    - name: Install packages
      apt:
        name: nginx
        state: present

    - name: Start service
      service:
        name: nginx
        state: started

    - name: Create file
      file:
        path: /tmp/test
        state: touch
```

Run playbook:

```
ansible-playbook site.yml
ansible-playbook site.yml -i hosts
ansible-playbook site.yml -u ubuntu
ansible-playbook site.yml -v          # Verbose
ansible-playbook site.yml -vvv       # Very verbose
ansible-playbook site.yml --check     # Dry run
ansible-playbook site.yml --syntax-check
```

Variables & Facts

Using variables:

```
---
- hosts: all
  vars:
    http_port: 80
    app_name: myapp

  tasks:
    - name: Print variables
      debug:
        msg: "App {{ app_name }} on port {{ http_port }}"

    - name: Use facts
      debug:
        msg: "OS is {{ ansible_os_family }}"
```

Variable files:

```
---  
- hosts: all  
  vars_files:  
    - vars/common.yml  
    - vars/{{ environment }}.yml  
  
  tasks:  
    - include_vars:  
      file: vars/additional.yml
```

Role Structure

Create role:

```
ansible-galaxy role init myrole

# Role structure
myrole/
  defaults/
    main.yml
  files/
  handlers/
    main.yml
  tasks/
    main.yml
  templates/
  vars/
    main.yml
```

Use roles:

```
---
- hosts: webserver
  roles:
    - common
    - nginx
    - { role: app, tags: "application" }

post_tasks:
  - name: Verify
    debug:
      msg: "Deployment complete"
```



Common Modules

Popular Modules

File operations:

```
- name: Create directory
  file:
    path: /tmp/mydir
    state: directory
    mode: '0755'

- name: Copy file
  copy:
    src: /local/path
    dest: /remote/path
    owner: root
    mode: '0644'

- name: Template file
  template:
    src: app.conf.j2
    dest: /etc/app/config
```

Package management:

```
- name: Install package
  apt:
    name: nginx
    state: present
    update_cache: yes

- name: Install multiple
  apt:
    name:
      - nginx
      - curl
      - git
    state: present

- name: Remove package
  apt:
    name: apache2
    state: absent
```

Service management:

```
- name: Start service
  service:
    name: nginx
    state: started
    enabled: yes

- name: Restart service
  service:
    name: nginx
    state: restarted

- name: Stop service
  service:
    name: nginx
    state: stopped
```



Control Flow

Conditionals & Loops

If conditions:

```
- name: Install package on Debian
  apt:
    name: nginx
    when: ansible_os_family == "Debian"

- name: Install on RedHat
  yum:
    name: nginx
    when: ansible_os_family == "RedHat"

- name: Check if installed
  debug:
    msg: "Installed"
    when: nginx_installed | default(false) | bool
```

Loops:

```
- name: Create users
  user:
    name: "{{ item }}"
    state: present
  loop:
    - alice
    - bob
    - charlie

- name: Install packages
  apt:
    name: "{{ item }}"
  loop:
    - nginx
    - curl
    - git
```

Handlers & Filters

| Advanced Features

Handlers (notify):

```
---
- hosts: all
  tasks:
    - name: Update config
      template:
        src: nginx.conf.j2
        dest: /etc/nginx/nginx.conf
        notify: restart nginx

  handlers:
    - name: restart nginx
      service:
        name: nginx
        state: restarted
```

Filters:

```
- name: Use filters
  debug:
    msg: "{{ variable | upper }}"

- debug:
    msg: "{{ variable | lower }}"

- debug:
    msg: "{{ list | join(', ') }}"

- debug:
    msg: "{{ dict | to_nice_json }}"
```

✓ Best Practices

✓ **Use roles for organization** - Structure code in reusable roles

✓ **Use variables for flexibility** - Make playbooks configurable

✓ **Test with --check** - Preview changes before executing

✓ **Use tags for selective runs** - Run specific parts of playbook

✓ **Leverage handlers** - Use notify for service restarts

Troubleshooting

🔍 Debug mode:

`ansible-playbook site.yml -vvv` - Very verbose output

`ansible-playbook site.yml --step` - Interactive

🔍 Test connection:

`ansible all -i hosts -m ping`

`ansible all -i hosts -m setup`

💡 Common Commands:

`ansible-inventory --list` - List inventory

`ansible-doc module_name` - Module documentation



Quick Reference

Command	Purpose	Example
<code>ansible</code>	Ad-hoc command	<code>ansible all -m ping</code>
<code>ansible-playbook</code>	Run playbook	<code>ansible-playbook site.yml</code>
<code>ansible-galaxy</code>	Manage roles	<code>ansible-galaxy role init</code>
<code>ansible-inventory</code>	Manage inventory	<code>ansible-inventory --list</code>
<code>ansible-doc</code>	Module docs	<code>ansible-doc apt</code>
<code>ansible-vault</code>	Encrypt data	<code>ansible-vault create secret.yml</code>
<code>--check</code>	Dry run	<code>ansible-playbook site.yml --check</code>
<code>-v, -vv, -vvv</code>	Verbosity	<code>ansible-playbook site.yml -vvv</code>



Ansible Concepts

File Modules

file

copy

template

lineinfile

fetch

System Modules

user

group

service

command

shell

Package Modules

apt

yum

pip

docker_container

Network Modules

uri

wait_for

get_url

ssh



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